

BENNINGTON MORSE, VT, USAWMO: **726166**Lat: **42.891N**Lon: **73.247W**Elev: **252**StdP: **98.34**Time Zone: **-5.00 (NAE)**Period: **01-19**WBAN: **54781****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-17.2	-25.8	0.4	-19.1	-22.9	0.5	-15.7	10.8	5.8	8.9	2.6	1.3	280	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.3	22.3	28.9	21.4	27.6	20.7	23.5	28.4	22.7	27.4	21.8	26.2	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.1	26.6	21.1	16.3	25.5	20.2	15.4	24.6	71.1	28.5	67.9	27.3	64.8	26.3	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.5	7.3	6.0	DB	-24.5	32.9	3.0	1.0	-26.7	33.6	-28.4	34.2	-30.1	34.7	-32.3	35.5
(5)				WB	-25.1	25.4	3.1	0.9	-27.3	26.1	-29.2	26.6	-30.9	27.1	-33.2	27.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.3	-3.8	0.8	7.9	13.8	18.0	20.8	19.8	16.0	9.8	3.9	-1.5
	DBStd	10.27	6.96	6.05	5.79	5.13	4.45	3.42	2.74	2.98	3.96	4.71	4.96	5.73
	HDD10.0	1881	474	388	292	98	14	0	0	0	2	63	191	358
	HDD18.3	3842	732	621	545	314	154	46	8	19	91	266	432	614
	CDD10.0	1304	1	1	5	37	131	240	336	304	183	58	9	2
	CDD18.3	225	0	0	0	2	12	36	86	66	22	2	0	0
	CDH23.3	2087	0	0	2	45	159	357	781	533	198	12	0	0
	CDH26.7	483	0	0	0	10	33	83	201	115	42	1	0	0
Wind	WSAvg	2.2	2.4	2.6	2.8	2.9	2.2	1.8	1.5	1.4	1.7	2.2	2.4	2.4
Precipitation	PrecAvg	1217	88	75	95	95	106	111	111	105	103	110	98	99
	PrecMax	1438	150	156	169	172	194	210	189	222	183	255	157	171
	PrecMin	928	42	45	44	31	36	44	49	31	52	10	63	40
	PrecStd	154	32	26	30	36	43	43	41	49	36	65	27	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	16.1	21.4	28.0	29.6	31.3	32.2	31.4	30.4	24.9	20.2	16.1
		MCWB	13.2	10.9	13.1	16.5	20.1	22.8	23.5	23.5	22.4	18.3	14.6	13.5
	2%	DB	10.6	11.4	15.8	23.9	27.3	29.0	30.3	29.2	27.6	22.6	17.4	12.3
		MCWB	7.5	6.8	10.4	14.0	18.7	21.2	22.8	21.9	20.7	17.3	12.9	9.9
	5%	DB	6.9	7.6	12.4	20.6	24.9	27.2	28.8	27.8	25.8	20.3	14.9	9.3
		MCWB	4.6	4.4	7.4	12.6	17.3	20.1	21.8	21.3	19.9	15.8	11.1	7.2
	10%	DB	4.1	4.8	9.7	17.5	22.6	25.3	27.5	26.4	23.5	17.8	12.4	6.9
		MCWB	2.0	2.2	5.8	10.8	16.1	18.7	21.1	20.5	18.4	14.0	9.4	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.5	13.9	18.6	21.7	23.6	25.1	24.5	23.3	20.2	15.9	13.7
		MCDB	15.6	15.7	18.6	24.5	27.9	29.3	30.4	29.4	28.4	23.4	19.0	16.0
	2%	WB	8.0	7.6	11.2	15.7	20.1	22.5	23.7	23.1	21.8	18.5	13.9	10.2
		MCDB	10.1	10.7	15.2	22.0	25.8	27.6	28.7	27.6	26.4	21.6	16.7	12.2
	5%	WB	4.8	4.6	8.2	13.7	18.6	21.3	22.8	22.3	20.5	16.5	11.8	7.5
		MCDB	6.8	7.5	11.4	19.7	23.4	25.7	27.5	26.5	24.3	19.8	14.3	9.3
	10%	WB	2.2	2.2	5.9	11.4	17.2	20.0	21.9	21.3	19.3	14.5	9.6	4.8
		MCDB	4.0	4.7	9.3	16.2	21.4	23.9	26.3	25.0	22.7	17.2	12.3	6.8
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.8	10.3	11.7	12.1	11.6	11.8	11.9	12.4	10.6	9.5	8.1
		MCDBR	12.0	12.6	14.6	17.2	15.8	14.1	13.3	13.0	14.2	13.9	13.0	11.1
		MCWBR	10.3	9.7	9.5	9.1	8.5	7.4	6.7	6.8	8.0	8.8	9.4	9.8
	5% WB	MCDBR	11.7	12.2	12.9	14.4	13.2	12.2	11.6	11.2	12.1	12.0	11.0	10.6
		MCWBR	10.3	10.0	9.6	8.6	7.8	6.9	6.3	6.3	7.1	8.4	9.2	9.7
Clear-Sky Solar Irradiance	taub	0.292	0.308	0.326	0.348	0.383	0.427	0.446	0.420	0.365	0.341	0.307	0.287	
	taud	2.374	2.319	2.385	2.413	2.355	2.250	2.206	2.263	2.402	2.480	2.525	2.474	
	Ebn at Noon	845	888	917	919	891	848	826	837	864	845	830	826	
	Edh at Noon	79	99	105	111	121	135	140	128	103	85	69	66	
All-Sky Solar Radiation	RadAvg	1.59	2.47	3.55	4.51	5.12	5.54	5.75	5.15	4.18	2.61	1.77	1.26	
	RadStd	0.14	0.23	0.28	0.50	0.56	0.49	0.40	0.27	0.34	0.28	0.18	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(